Residential Water System - Optimized Concept

Yamoah Residence | 36 ft x 36 ft plot | Four tanks A-D | Two existing pumps | Nine mapped outlets

Central proposal: give municipal water first priority at the outlets without filtration; when municipal supply cannot sustain service, transfer the **entire outlet network** to water that has passed through the **one shared filter**. Retain the existing four storage tanks and pumps primarily for replenishment/transfer. Install a physically appropriate, approved positively isolating changeover arrangement. All red tank-source pipes must terminate at the filter or another tank - never directly at a tap.

1. Site interpretation

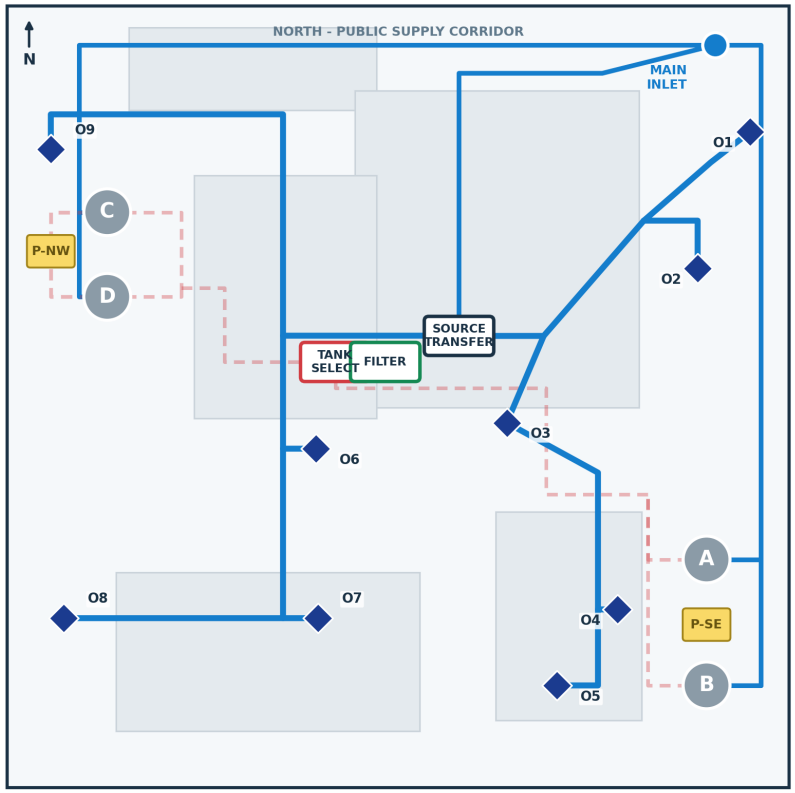
Location / component	Observed / provided arrangement	Design implication
NE: municipal inlet	One incoming mains connection, intermittent Tue AM and Thu PM	One blue service manifold for direct fixtures and protected tank fills; test tap stays blue.
SE: tanks A and B	A outlet +3.7 m / inlet +5.0 m; B outlet +7.2 m / inlet +9.5 m; P-SE near A	B gravity is strongest tank source; P-SE retains emergency A/B-to-C transfer role only if needed.
NW: tanks C and D	C outlet +0.6 m / inlet +2.6 m; D outlet +4.2 m / inlet +6.2 m; P-NW near C	C is low-pressure reserve / collection; P-NW moves C to elevated D only when needed.
Nine outlets O1-O9	Scattered at plot perimeter/interior according to source sketch	A single shared outlet distribution tree after source transfer avoids two pipe sets to every tap.

Important: site-plan footprints and pipe routes in the following diagrams are reconstructed from the linked sketch, not traced construction measurements. Unknown concealed connections, tank volumes, pump models, pipe diameters and actual outlet elevations must be surveyed before work starts.

2. Proposed layout - municipal water available

YAMOAH RESIDENCE | CONCEPT PIPING LAYOUT

Single filter / shared outlet distribution / mains-first operation



Plot approx. 36 ft x 36 ft | Outlet positions reproduced schematically from supplied sketch

OPERATING MODE

MUNICIPAL SUPPLY ON

Blue mains feed the outlets directly. Tank-filling branches remain available through protected air gaps. No filtration of municipal water.

- Municipal supply pipe
- Tank water / transfer
- Filtered tank-water pipe
- Existing outlet O1-O9

CRITICAL SAFETY POINT

An approved, positively isolating source-transfer assembly must prevent tank water entering the municipal system. Tank inlets need air gaps and float control.

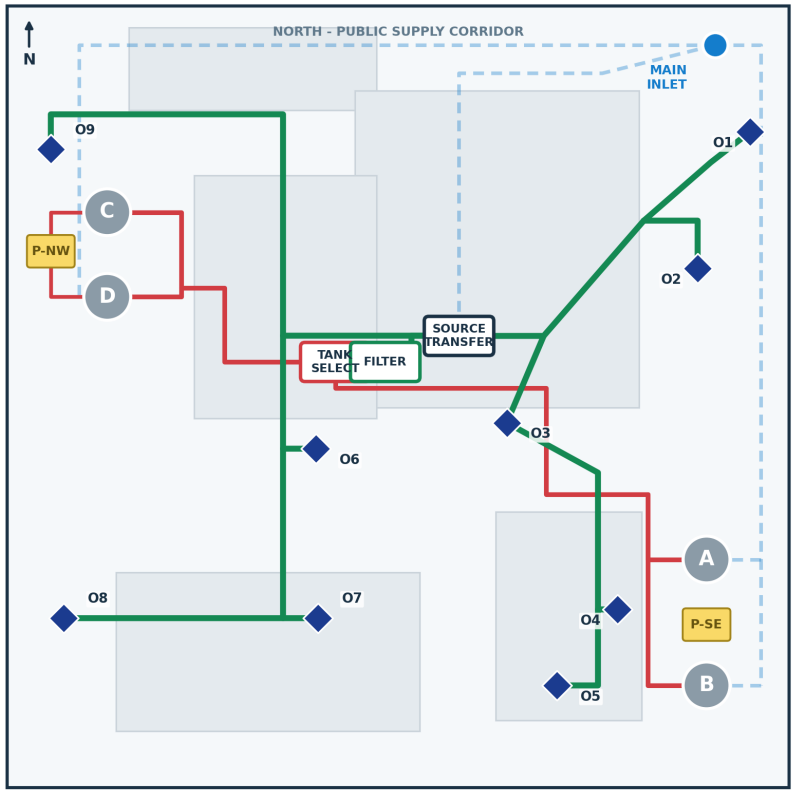
CONCEPT ONLY • Routing diagram is not to scale; verify pipe accessibility, heights, hydraulic head, valve types and actual connections onsite.

Blue outlet distribution represents the installed shared pipes while the municipal port is selected. The faint red/green pipes are installed but inactive. All municipal tank-filling connections bypass the filter and require approved air-gap/float arrangements.

3. Proposed layout - municipal supply unavailable

YAMOAH RESIDENCE | CONCEPT PIPING LAYOUT

Single filter / shared outlet distribution / mains-first operation



Plot approx. 36 ft x 36 ft | Outlet positions reproduced schematically from supplied sketch

OPERATING MODE

MUNICIPAL SUPPLY OFF

Tank outlets feed the common filter through a red, isolated source manifold. Only green filtered water reaches outlets. Blue municipal pipes are inactive.

- Municipal supply pipe
- Tank water / transfer
- Filtered tank-water pipe
- Existing outlet O1-O9

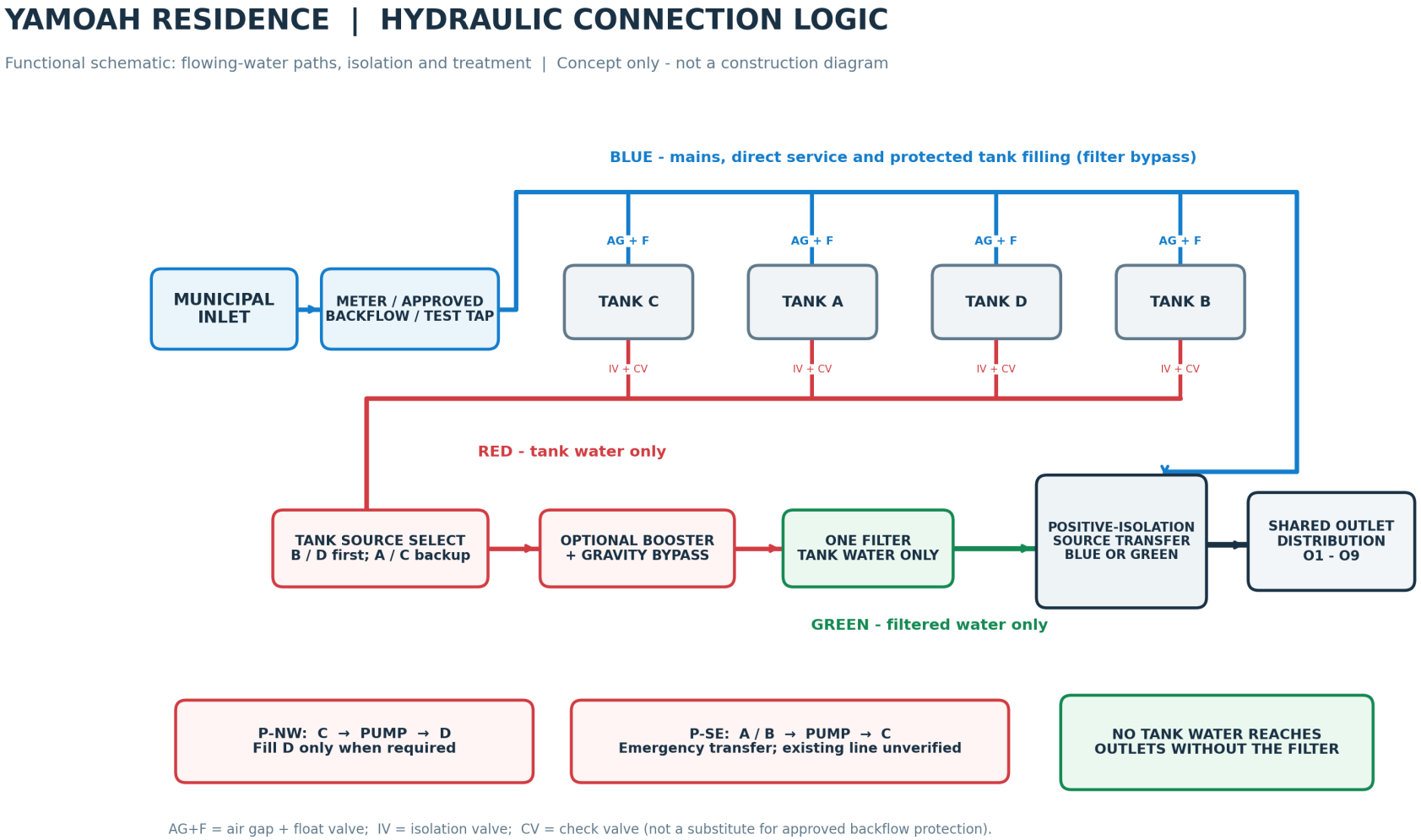
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An approved, positively isolating source-transfer assembly must prevent tank water entering the municipal system. Tank inlets need air gaps and float control.

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The same physical outlet-distribution pipes now carry green filtered storage water. The upstream red paths carry raw tank water only as far as the source manifold/filter; blue municipal piping is isolated and inactive.

4. Connection logic and plumbing boundaries



Blue fill and direct-service paths do not enter the filter. Independent tank takeoffs use isolation and directional protection; the source selector and changeover valve must not create an unprotected municipal-to-storage cross-connection.

5. Why gravity helps - and its limits

The static head below assumes a ground-level tap at elevation 0 m, a full tank water surface approximately near the stated inlet elevation, and no pipe/filter/valve losses. Real pressure at elevated fixtures will be lower.

Tank	Outlet level / full-water estimate	Approx. static head at ground tap	Assessment
A	+3.7 to +5.0 m	0.36 to 0.49 bar	Marginal gravity after a whole-house filter; retain as emergency water source.
B	+7.2 to +9.5 m	0.71 to 0.93 bar	Best gravity source; test filter loss and highest served fixture.
C	+0.6 to +2.6 m	0.06 to 0.26 bar	Unsuitable for reliable direct service; pump or raise water to D.
D	+4.2 to +6.2 m	0.41 to 0.61 bar	Possible low-level gravity use with low-resistance filter; pressure verification essential.

Principle: 1 m water head is approximately 0.098 bar. Pressure comes from the tank **water-surface height**, not simply the vertical distance to the outlet pipe. Filter drop, route length and demand reduce residual pressure.

Potential pump-saving opportunities

- **B to C:** gravity transfer is plausible because the minimum B water surface (+7.2 m) exceeds the C inlet (+2.6 m). A/B-to-C transfer must remain isolated from the outlet network and end at a float-controlled, air-gapped inlet.
- **B to D:** gravity might also fill D (+6.2 m inlet), but the minimum theoretical margin is only around 1.0 m and friction can consume it. Treat as conditional, not guaranteed.
- **A to C:** minimum theoretical margin is about 1.1 m. Low flow is likely; use P-SE only when measured head/flow is insufficient. **A to D** is not reliably gravity-feasible.
- **C to D:** retain P-NW with a low-level cutout at C and high-level stop at D. Do not infer D is full merely because its pump stopped; a dry-run trip or power loss can look the same.

Municipal fill challenge: D (+6.2 m) and especially B (+9.5 m) require corresponding supply head plus friction losses. Observe city pressure under flow before relying on direct filling. Fill lower C and A when pressure is weak; elevate C-to-D via P-NW as needed.

6. Recommended pipework and controls

Circuit	Proposed routing	Required controls / cautions
BLUE - municipal	Meter/backflow/test tap at NE; one inlet manifold feeds a municipal port of the source selector and protected A/B/C/D fill branches.	Air-gap + float valve + screened overflow for each tank; pressure gauge and isolation. No blue pipe through filter.
RED - storage	Local A/B and C/D takeoffs each join a tank-only source header to an accessible central filter station. Retain existing P-SE to C and P-NW C to D transfer lines where sound.	Tank-by-tank isolation, non-return on each draw, level protection, strainer; avoid high tanks backfilling low tanks unintentionally.
GREEN - filtered	One filter at accessible central station; its outlet feeds stored-water port of the positively isolating transfer. One shared downstream tree serves all nine outlets.	Whole-house filter rated for measured peak flow and low pressure loss; sample points before and after; change and sanitize per manufacturer.
SOURCE TRANSFER	One common distribution network AFTER a mains/storage selector; municipal preferential while pressure/flow are adequate.	Engineer-approved backflow protection, positive isolation, no simultaneous connection; manual lockout and fail-safe behavior.

Equipment and sizing notes

- Start pipe sizing from simultaneous fixture demand, lengths, height losses and municipal pressure recordings. A 25-32 mm trunk, 20-25 mm zone feeds and 15-20 mm fixture branches are **discussion starting points only**, not a verified schedule.
- Locate the filter at an accessible, protected point where red tank sources converge. Include full-bore isolation valves, flushing/sample ports and pressure gauges. A service bypass must never send unfiltered red water to outlets.
- If gravity produces weak service after filtration, add a demand-controlled booster on the **red tank-water side before the filter**, with an appropriately arranged gravity bypass and potable-water-safe pressure controls; do not connect pumps to municipal suction.

7. Operating sequence, commissioning and safeguards

Municipal return - usually Tuesday AM / Thursday PM

- Check blue test tap and a pressure/flow sensor. When the municipal source can sustain a real outlet demand, transfer to **municipal-direct** service. Avoid switching on pressure alone if pressure is static but water is not flowing.
- Serve taps first. Fill storage via individually controlled float-valve inlets using available remaining flow. If pressure cannot reach elevated B or D, use the low tanks and scheduled pumps instead of aggressive municipal pumping.

Municipal supply drops

- Following a verified low-pressure/low-flow delay, isolate municipal port, select tank supply through the filter, then supply all outlets through green filtered distribution; apply switch-back hysteresis.
- Prefer B/D head for gravity where proven, with A/C reserve and booster only as required. Trigger P-NW using low D and adequate C levels; use P-SE only for verified SE-to-NW emergency replenishment.

Minimum on-site verification before cutting pipes

- Trace the actual hidden pipes from A, B, C, D and both pump discharges; map/check every isolation and check valve. Verify whether P-SE feeds C, D or both and whether P-NW stops due to a genuine level switch.
- Record elevations, tank capacities, pump curves, filter model/rated flow, all outlet heights, municipal pressure/flow during actual Tuesday and Thursday supply windows, and current static/flow pressure.
- Pressure-test, flush, clean/disinfect and sample as appropriate. Confirm no outlet can receive raw storage water, no reverse flow can reach municipal supply, no gravity backfill of adjacent tanks, no dry-run and no unsupported elevated tank loads.

Safety, approvals and drinking-water quality

Use a Ghana-qualified plumber / hydraulic designer to validate the arrangement against Ghana Building Code GS 1207:2018 and the local water utility requirements. A normal check valve alone is not an acceptable substitute for required cross-connection protection. Tank inlets require robust air gaps and overflows; all tank-water delivered to taps must pass through filtration. Filtration **does not automatically mean pathogen-safe drinking water**; specify the filter against laboratory water testing and verify disinfection if needed.

References: user-provided site sketch; WHO, *Safe Piped Water* (air gaps / break tanks); American Water Works Association, *Cross Connection* policy; WHO, *Sanitary Inspection Package - Piped Distribution: Storage Tank* (2026); Ghana Building Code GS 1207:2018, Part 30 (water supply and backflow).